



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Final Exam Spring 2024

CSE 2233/CSI 233: Theory of Computation/Theory of Computing

Total Marks: 40

Duration: 2 Hours

Answer all questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

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1. Consider the following Context-free grammars (CFG) and answer according to it: 4*2
- a) $S \rightarrow AB \mid C$
 $A \rightarrow aAb \mid ab$
 $B \rightarrow cBd \mid cd$
 $C \rightarrow aCd \mid aDd$
 $D \rightarrow bDc \mid bc$ With the help of **Top-Down Parse Trees**, find out if the grammar is Ambiguous or not for the string “**aabbccdd**”.
- b) $E \rightarrow +E \mid -E \mid *E \mid X$
 $X \rightarrow pXq \mid pYq$
 $Y \rightarrow rYw \mid rw$ With the help of **Leftmost derivation**, derive the following string “**+*-pppprrwwqqq**”.
2. Design CFGs that generate the following languages: 2*4
- a) Design a CFG for a language of all strings over $\{1,0\}$ that contains “**110**”, “**010**” or “**011**”.
- b) Design a CFG for a language of all strings over $\{1,0\}$ representing binary numbers divisible by 4.
- c) Design a CFG for $L = \{ X^{2n} (Y^m Z^m)^n \mid m,n \geq 0 \}$ over $\{X,Y,Z\}$
(e.g. XXXXYZZYZZ, XXYYZZZZ)
- d) Design a CFG for $L = \{ a^n b^m c^p \mid \text{where, } (n,m > 1), (n \neq m) \text{ and } (p > 3) \}$ over $\{a,b,c\}$
3. Convert the following grammars into Chomsky Normal Form (CNF): 4*3
- a) $E \rightarrow T \mid E + T$
 $T \rightarrow F \mid T * F$
 $F \rightarrow I \mid (E)$
 $I \rightarrow a \mid b \mid Ia \mid Ib \mid IO \mid II$
- b) $S \rightarrow ASB \mid \epsilon$
 $A \rightarrow aAS \mid a$
 $B \rightarrow SbS \mid A \mid bb$
- C) $A \rightarrow BAB \mid B \mid \epsilon$
 $B \rightarrow 00 \mid \epsilon$

4. Draw the Push Down Automata (PDA) for the following languages:

4*2

a) $L = \{ a^m b^n c^n a^m \mid m, n > 0 \}$

b) $L = \{ w@w^r \mid w \text{ is a string over } \{a, b\}^* \text{ and } w^r \text{ is the reverse of } w \}$

5. Draw a Turing Machine for the following language.

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$$L = \{ a^i b^j c^k \mid i < j < k, i \geq 1 \}$$